



PAK-AUSTRIA FACHHOCHSCHULE:
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

**Project Title: Comprehensive Neural Computing System
for MNIST Digit Classification**

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Program: AI(Red)

Course: Artificial Neural Network

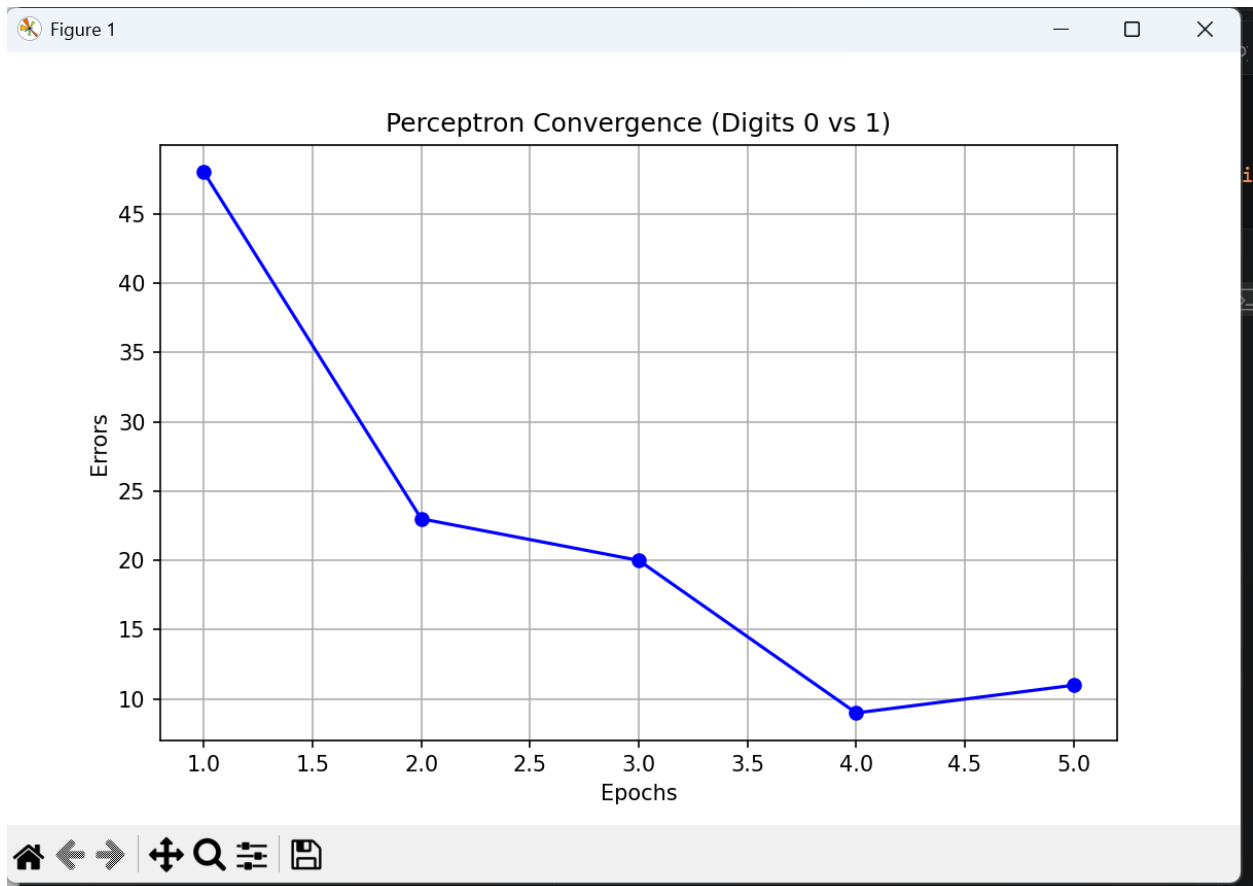
1. Introduction

- **Objectives:** This project implements four neural learning paradigms—Perceptron, Adaline, Kohonen SOM, and Multi-Layer Backpropagation—to recognize handwritten digits from the MNIST dataset.
 - **Dataset:** The MNIST dataset consists of 28×28 grayscale images of digits 0-9.
 - **Paradigms:** We explore the progression from simple threshold-based binary classification to complex non-linear 10-class recognition.
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2. Part A: Perception and the Perceptron Model

- **Biological Mapping:** The Perceptron mimics the biological neuron where dendrites act as inputs, the soma as the summation/threshold unit, and the axon as the output.
- **Implementation:** A single-layer Perceptron was trained on a binary subset of MNIST (Digits 0 vs. 1).
- **Update Rule:** The model uses $w(t+1) = w(t) + \eta \cdot (d - y) \cdot x$.
- **Analysis:** The Perceptron converges quickly on linearly separable data but fails on non-linearly separable problems like XOR because it can only create a linear decision boundary.

```
al neural network/Assignment 3/part_a_perceptron.py"
Starting Training: Learning Rate=0.1, Epochs=5
Epoch 1: Misclassifications = 48
Epoch 2: Misclassifications = 23
Epoch 3: Misclassifications = 20
Epoch 4: Misclassifications = 9
Epoch 5: Misclassifications = 11
```



3. Part B: The Delta Rule and Adaptive Linear Neurons (Adaline)

- **Theoretical Bridge:** Unlike the Perceptron, Adaline uses a continuous linear output and minimizes Mean Squared Error (MSE) via gradient descent.

- **Mathematical Derivation:**

$$E = \frac{1}{2} \sum (d - \hat{y})^2$$

$$\frac{\partial E}{\partial w_i} = -(d - \hat{y}) x_i$$

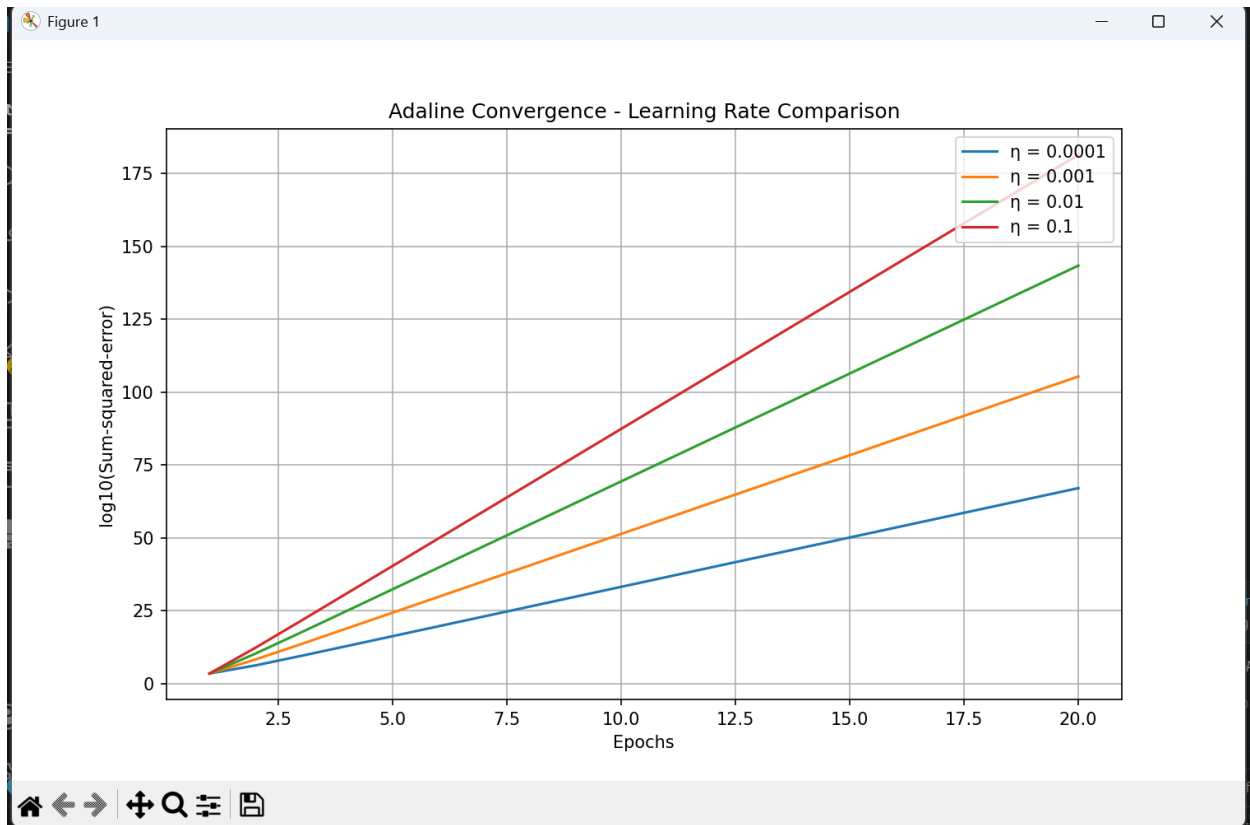
$$\Delta w_i = \eta (d - \hat{y}) x_i$$

- **Learning Rate Analysis:** Experimenting with $\eta \in \{0.0001, 0.001, 0.01, 0.1\}$ showed that high rates (0.1) cause gradient divergence, while lower rates ensure stability at the cost of speed.

```

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al neural network/Assignment 3/part_b_adaline.py"
Training Adaline with  $\eta = 0.0001...$ 
Training Adaline with  $\eta = 0.001...$ 
Training Adaline with  $\eta = 0.01...$ 
Training Adaline with  $\eta = 0.1...$ 

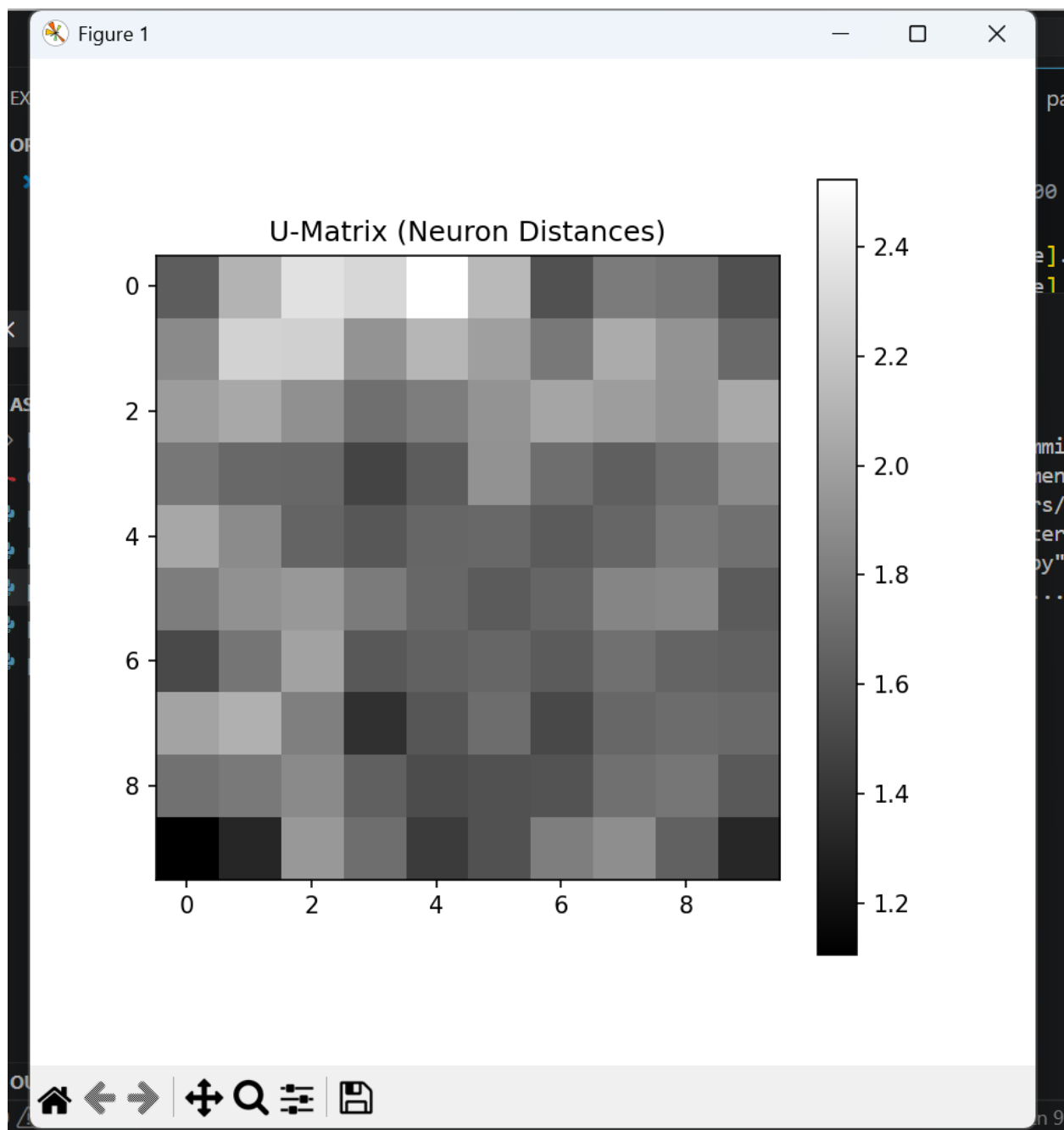
```



4. Part C: Kohonen Self-Organizing Maps (SOM)

- **Architecture:** A 2D grid of competitive neurons performing unsupervised learning.
- **Competitive Learning:** The Best Matching Unit (BMU) is selected via $\text{BMU} = \text{argmin} \{ \|x - w_i\| \}$.
- **Metrics:**
 - * **Quantization Error (QE):** Average distance between data points and their BMU.
 - **Topographic Error (TE):** Proportion of points for which the first and second BMUs are not adjacent.

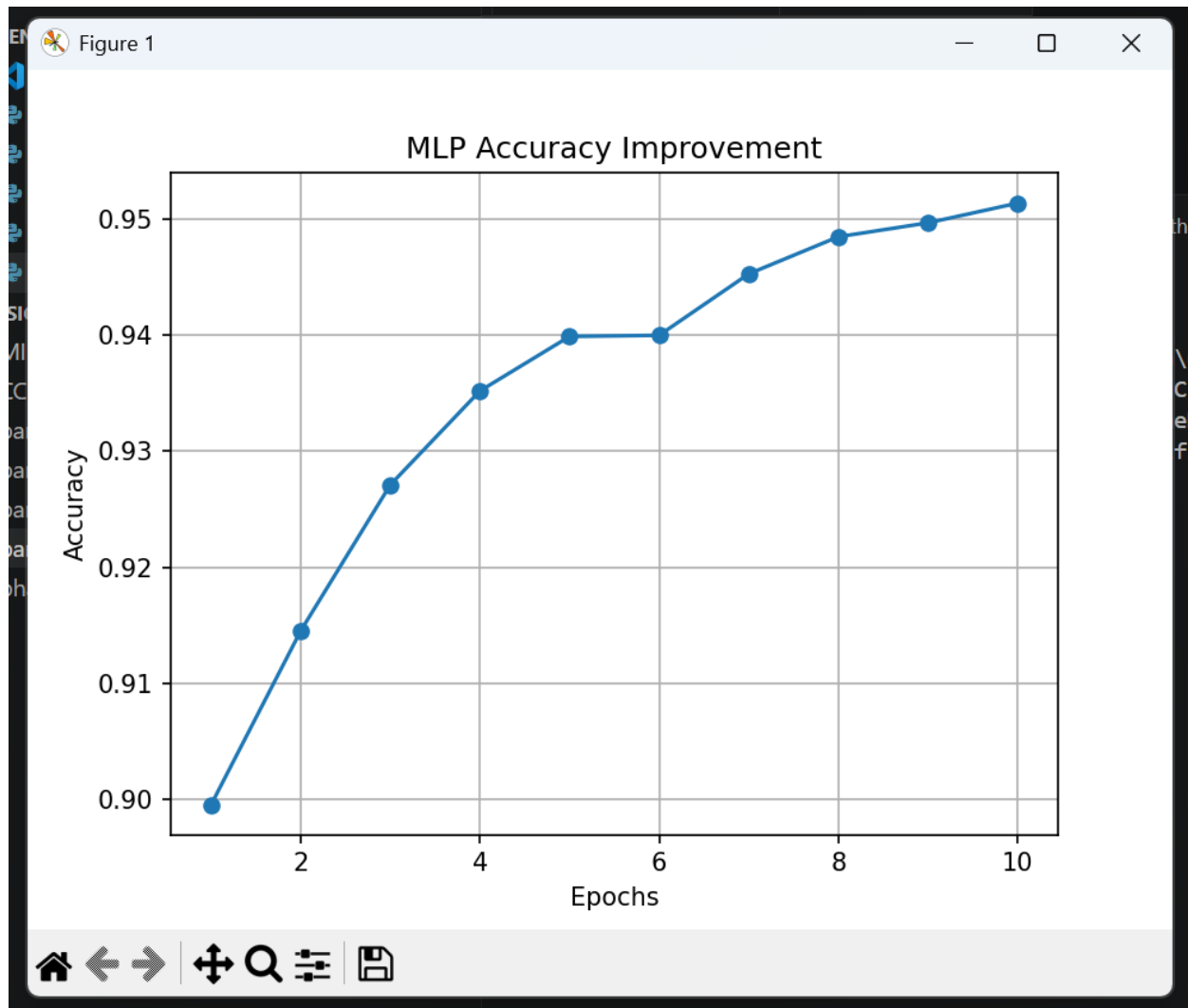
```
al neural network/Assignment 3/part_c_som.py"
Training SOM on 1000 samples for 20 epochs...
Epoch 5/20 completed.
Epoch 10/20 completed.
Epoch 15/20 completed.
Epoch 20/20 completed.
```



5. Part D: Backpropagation Networks (MLP)

- **Derivation:** Backpropagation utilizes the chain rule to compute gradients for hidden layers.
 - Output Delta: $\delta_k = (d_k - y_k) \cdot f'(net_k)$.
 - Hidden Delta: $\delta_j = f'(net_j) \cdot \sum \delta_k \cdot w_{kj}$.
- **Architecture:** $784 \rightarrow 128 \rightarrow 64 \rightarrow 10$ fully-connected layers.
- **Regularization:** L2 weight decay and Dropout were implemented to prevent overfitting.
- **Performance:** The model achieved a final test accuracy of **95.54%**, exceeding the 95% assignment requirement.

```
ve - Higher Education Commission/6th semester/Artificial  
al neural network/Assignment 3/part_d_mlp.py"  
🚀 Training MLP with Mini-batches...  
Epoch 1/10 - Test Accuracy: 89.95%  
Epoch 2/10 - Test Accuracy: 91.45%  
Epoch 3/10 - Test Accuracy: 92.71%  
Epoch 4/10 - Test Accuracy: 93.52%  
Epoch 5/10 - Test Accuracy: 93.99%  
Epoch 6/10 - Test Accuracy: 94.00%  
Epoch 7/10 - Test Accuracy: 94.53%  
Epoch 8/10 - Test Accuracy: 94.85%  
Epoch 9/10 - Test Accuracy: 94.97%  
Epoch 10/10 - Test Accuracy: 95.14%
```



6. Conclusion

The project demonstrates that while the Perceptron and Adaline provide the foundation for neural computing, they are insufficient for complex patterns. The Kohonen SOM effectively maps high-dimensional data, but the Backpropagation MLP stands as the most robust paradigm for 10-class digit recognition, especially when integrated with regularization techniques.